Supporting Teachers' Data-Based Instruction in Early Writing: Tools, Learning, and Collaborative Support

Grant #R324A130144

Accomplishments

A. What are the major goals of the project?

As described in the performance agreement, the major goals of the project are to:

(1) Develop a professional development system (DBI-TLC) to support teachers' use of data-based instruction with students with or at risk for disabilities that affect early writing development, and
(2) Assess the feasibility and potential for efficacy of the intervention for improving student writing outcomes.

B. What was accomplished under these goals?

The status of the major activities of the project for Year 1 (September 1, 2013-August 31, 2014) are listed in Table 1, and described in more detail in the narrative below.

Table 1
Project Activities and Status

Project Activity (As listed in Performance Agreement)	Status
1. Identify implementers.	In progress
2. Develop and revise the DBI protocol (Manual with processes and tools).	In progress
3. Develop fidelity of implementation instrument.	In progress
4. Develop learning modules for DBI-TLC.	In progress
5. Develop teacher knowledge and skills assessment.	To begin in April
6. Host summer institute for implementers.	To be conducted in July-August

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1) Identify implementers. We have met with our School Partners in Minneapolis (MN) and Columbia (MO) Public Schools to review the project goals and timeline, and to identify participants for the Summer Institute to be held in Summer, 2014. School Partners have agreed to assist us in identifying 6 to 8 teachers per site who provide direct services to students in Grades 1-3 identified as at-risk or as having disabilities that affect their early writing skills. Participating teachers will be identified and recruited for the Summer Institute by the end of the 2013-14 school year.

2) **Develop and revise the DBI protocol (Manual with processes and tools).** We spent the first half of Year 1 developing the DBI Tools listed in Table 2 (taken from the original Table 1 of the grant proposal). Specifically, our research teams divided into 4 groups (with each group consisting of representatives from both MN and MO) to complete development activities within each major component of DBI (listed in the left column of Table 2). These activities are described in more detail below.

Table 2
DBI Protocol Components and Status

DBI Component	Tools	Status
DBI Overview	1. DBI Manual 2. AIRS fidelity form	1. Manual drafted. 2. Manual sent to Content Experts Deno & Fuchs for feedback. 3. Manual revisions completed. 4. Manual sent to School Partners for feedback. 5. Manual revisions in progress. 6. AIRS fidelity form under revision.
Assessing Student Performance and Progress using CBM	1. CBM probes 2. Administration directions 3. Scoring guidelines 4. Graphic procedures 5. Procedures for interrater agreement (IRA)	1. CBM probes revised, refined, and piloted. 2. Administration directions revised, refined, & piloted. 3. Scoring guidelines piloted and under revision. 4. Graphing procedures under development. 5. Procedures for assessing IRA developed and piloted.
Data-Based Decision Making	Step-by-step procedures for: 1. Setting long-range goals. 2. Applying decision rules for instructional changes. 3. Decision-making rubric 4. Changes in Instruction Plans (CIP) form	1. Literature review completed. 2. Findings from literature review are being incorporated into DBI manual (including procedures for setting goals, making instructional decisions, and implementing instructional changes using CIP).
Selecting and implementing interventions	1. Research-based options for Writing Instruction 2. Basic Instructional Plan (BIP) 3. Guidelines for individualizing instruction	1. Literature review completed. 2. Currently exploring templates for basic instruction plan (BIP) for writing. 3. Guidelines for individualizing instruction in progress.

DBI overview. A primary objective of Year 1 has been to draft a manual of the overall DBI process. We conducted this process in the following steps:

a) We held a telephone conference with our research team (both in MN and MO), Content Experts Stan Deno and Lynn Fuchs, and School Partners Ann Casey (Minneapolis Public Schools; MPS) and Lou Ann Tanner-Jones (Columbia Public Schools; CPS). During this call, we reviewed our overall plan and asked the Experts and Partners to comment on:

- Their general impressions of our proposed DBI process;
- Challenges we might anticipate in developing a DBI process that is feasible and sustainable for teachers to implement;
- What resources to consider (in addition to those already identified); and
- Any additional advice.
- We summarized notes from the conference call and discussed the major themes within our research team. Most of the discussion centered around how to make the DBI process manageable and feasible for teachers to implement, as well as the importance of providing specific guidance related to how to use the data to make instructional decisions. The School Partners also emphasized the importance of providing guidance about how to link DBI to general classroom instruction. This initial feedback has informed many of the components included in the current DBI manual and tools.

b) We collected existing manuals related to CBM and DBI, including the original *Data-Based Program Modification* manual (Deno & Mirkin, 1977), *Using CBM for Progress Monitoring in Written Expression and Spelling*, (L.S. Fuchs & D. Fuchs, 2007), and materials from the National Center on Intensive Intervention (NCII; http://www.intensiveintervention.org/). We used these resources to outline the major components and draft the current DBI manual.

c) After drafting and reviewing the DBI manual within our research team, we forwarded the manual to Content Experts Stan Deno and Lynn Fuchs, and School Partners Ann Casey and Lou Ann Tanner-Jones. We asked them to provide feedback in the form of tracked changes, as well as to comment on its overall *comprehensiveness*, *organization*, *clarity*, *accuracy*, and *utility for teachers*. Content Experts sent us feedback via email. School Partners provided feedback via face-to-face meetings at our respective sites.

d) We met with Content Expert Steve Graham (while attending a conference) to describe the overall project and get his input about the best way to provide teachers with diagnostic feedback and instructional guidance related to beginning writing.

e) We incorporated all of the above feedback and guidance from Content Experts and School Partners into revisions of the DBI manual. The current draft is provided in Appendix A.

Assessing student performance and progress using CBM. The CBM group completed the following tasks:

a) We determined that additional Word Dictation probes were needed to have a sufficient number of prompts for progress monitoring. Thus, we created procedures for generating Word Dictation probes:
   - Designed Word Dictation lists to provide diagnostic feedback on students' spelling by including patterned words and irregular words across probes.
   - Patterned word types selected from the Common Core standards for grades 1 and 2. Word patterns shared across grades comprised larger percentages of Word Dictation stimuli (e.g., CVC, CVCe, CCVCe words); conversely, word patterns common only

to one grade comprised smaller percentages of Word Dictation stimuli (e.g., CCVC, CVVC).

- Reviewed the literature for word lists containing patterned word types.
- Selected patterned words for Word Dictation from the following lists: 5th edition of Direct Instruction Reading (Carnine, Silbert, Kame'enui & Tarver, 2008) and The Basic Spelling Vocabulary List (Graham, Harris & Loynachan, 1993).
- Refined word selection by sampling across common blends and digraphs within lists of patterned word types.
- Created a database of words selected for each word type.
- Developed a randomization process for distributing words across Word Dictation probes: words within patterned word types were randomly selected and randomly ordered within each Word Dictation prompt.

b) Developed CBM-W probes for piloting, using existing probes but combining them into probes of appropriate lengths to administer with students at different grade levels.
   - Produced 4 distinct Word Dictation prompts each containing 40 words.
   - Produced 4 Picture Word prompts each containing 24 pictures. Picture Word prompts were previously made and rearranged to produce non-overlapping combinations of prompts.
   - Produced 4 Story Prompts each containing 1 unique story starter. Story Prompts were previously made and rearranged in non-overlapping combinations.

c) Revised CBM-W administration and scoring procedures (Revised scoring procedures are in Appendix B)
   - Updated CBM-W administration directions to include new prompt examples and practice times.
   - Updated CBM-W scoring procedures with refinements in the areas of capitalization, punctuation, verb tense, words spelled correctly, and correct letter sequences.

Data-based decision making. In the area of Data-Based Decision Making (DBDM), we completed the following tasks:

a) Reviewed literature resources already gathered from the grant proposal and conducted a literature search to find additional articles related to DBDM. We searched for articles related to all areas of CBM, including reading, math, and writing. We focused on research that would assist us in determining effective decision-making rules, goal setting methods, and components needed for significant changes in student performance. We also researched the area of qualitative indicators of writing performance.

b) Reviewed existing manuals and other materials that have rubrics for DBDM and Change in Instructional Plans (CIPs).

c) Found and compiled into an Excel file all articles located through the literature review that were relevant to data-based decision making. We reviewed and summarized each article in the following format: APA citation, brief summary with main points (e.g., important findings, type of CBM, length of study, decision rules used, outcomes measured) and then a more in-depth summary with additional details of study outcomes.

d) Created a draft Decision Making Rubric (see Appendix C).

e) Created a draft Change in Instructional Plan form (see Appendix D).

f) Created an Executive Summary (see Table 3) to summarize key ideas and findings from our literature review. This summary included recommendations gleaned and further questions to consider for the following areas: data decision rules (trend line vs. 4-point rule); number of data points needed to make good instructional decisions; effectiveness of CBM implementation in improving student outcomes; goal setting; and types of effective feedback for teachers (computerized, expert consultant, teacher self-regulation). The chart also included questions raised from our research summaries about the use of qualitative indicators of writing performance as a component of progress monitoring. Recommendations included in this Executive Summary informed procedures included in the DBI manual and accompanying tools.

Table 3
Executive Summary of Data-Based Decision-Making Literature

Key Idea	What the Literature Says		Overall Recommendations and Questions to Consider
Making data-based decisions	4 point Rule: • Fuchs, Fuchs, Hamlett, & Ferguson (1992 *R*) growth monitored every 4 data points; trend line rule used if no clear decision after 8 data points • Fuchs, Fuchs, Hamlett, & Stecker (1991 *M*) – 4 consecutive scores below goal line implement change; if rule didn't prompt decision after 8 scores applied trend line rule • Fuchs, Fuchs, Hamlett & Allinder (1991a *S*) – used 4 pt. rule initially – if no clear decision after 8 pts switched to trend line	Trend Line & Data Points 4-6 Data Points • Fuchs, Fuchs, Hamlett, & Ferguson (1992 *R*) • Fuchs, Fuchs, Hamlett, & Stecker (1991 *M*) 7-10 Data Points • Wesson, Skiba, Sevcik, King, & Deno (1984 *R*) – • Wesson, Deno, Mirkin, Maruyama, Skiba, King, & Sevcik. (1988 *R*) • Fuchs, Fuchs, & Hamlett (1989a *R*) • Fuchs, Fuchs, & Hamlett (1989c *M*) • Fuchs, Fuchs, Hamlett (1989b *R*) • Fuchs, Deno, Mirkin (1984 *R*) – • Fuchs, Fuchs, Hamlett, & Allinder (1991b *S*) • Stecker & Fuchs (2000). • Fuchs, Fuchs, Hamlett, Stecker (1990 *M*) • Ardoin et. al. (2013 *R*) – • McMaster et al. (2011 *W*) >10 Data Points • VanNorman et al. (2013) • Christ et al. (2012 *R)* Other • Christ et al. (2013 *R*) – need dense data collection and longer data collection; accuracy improves with time	<u>Question:</u> Are a certain number of data points needed to obtain a reliable trend line or slope? <u>*Recommendation:*</u> *The literature seems to suggest use of a trend line rule with at least 7-10 data points.*
Progress monitoring	• Fuchs & Fuchs (2007) – use at least 5 weeks' worth of weekly progress monitoring (practitioner article) • See also Christ et al. (2013 *R*) cited above • Christ et al. (2012 *R)* at least 14 data points needed when collecting one per week – for reading		<u>*Recommendation:*</u> *At least 5 weeks of progress monitoring, but accuracy improves with time.*

Key Idea	What the Literature Says		Overall Recommendations and Questions to Consider
CBM implementation	CBM Alone: in general, CBM proven to be effective • Fuchs, Fuchs, Powell, Seethaler, Cirino, & Fletcher (2008 *M*) – CBM teachers design more varied instructional programs responsive to individual needs • Wesson, Skiba, Sevcik, King, & Deno (1984) – teachers using technically adequate CBM and evaluation criteria made better gains • Fuchs, Fuchs, Hamlet (1989a *S*) Spelling achievement significantly higher for SPED students with enhanced CBM feedback vs. control • Fuchs, Fuchs, Hamlett, Stecker (1990 *M*) CBM analysis resulted in significantly higher student achievement than control group for SPED students • Fuchs, Deno, Mirkin (1984 *R*) CBM teachers changed goals more and stated goals more specific than control teachers. Students in treatment were more aware of goals and progress and showed greater reading achievement than those in control condition. • Fuchs, Fuchs, Hamlett & Allinder (1991a *S*) CBM teachers compared to control increased student goals for 90%, made instructional adjustments an average of 2.74 times compared to control group of .17 adjustments • Allinder & Oats (2007 *M*) Teachers who perceive CBM as "acceptable" implement some components with greater fidelity – do probes more often and set ambitious goals.	CBM + Feedback: Computerized/Diagnostic Feedback • Fuchs, Fuchs, Hamlet (1989a *S*) Computerized teacher feedback (computer graphs results & provides decision rules) results in better compliance in using CBM results to make instructional modifications/goal changes. • Fuchs, Fuchs, & Hamlett (1989d *R*) CBM measurement and evaluation (CBM-M+E) resulted in more reliable gains in reading than control group, whereas CBM-M alone did not show significant differences. • Fuchs, Fuchs, Hamlett, & Allinder (1991b *S*) CBM with computer skill analysis resulted in more specific instructional changes (teachers) and showed more gains in spelling (students). • Capizzi & Fuchs (2005 *M*) – CBM + Diagnostic Feedback provided more targeted instruction to meet individual needs and targeted more appropriate skills • Fuchs, Fuchs, Hamlett, & Stecker (1991 *M*) – students in teacher groups receiving expert system instruction (CBM-ExS; computer programs attempting to reproduce advice provided by expert) had greater achievement and instructional gains • Fuchs, Fuchs, Hamlett & Allinder (1991a *S*) For CBM with SPED students, review of computer system feedback is sufficient to help teachers revise instruction (additional consultation was not needed) Expert Consultant/Teacher/Group Feedback • Wesson (1991 *R*) – CBM with group consultation shows greatest gains • Fuchs, Fuchs, Hamlett, Stecker (1990 *M*) Use of typical performance goal (average EOY score) plus skills analysis resulted in more specific instructional changes for SPED students • Fuchs, Fuchs, Hamlett (1989b *R*) Performance Indicator plus diagnostic analysis (PI+D) group wrote more specific instructional plans and effected better growth with their students.	<u>Question:</u> Is there a way to explicitly provide feedback with scripted and detailed examples in our manual that teachers can refer to? In a way, if we are first instructing a very small cohort of teachers, could they serve as an expert group, as resources to teachers who join in Year 3 and wherever this goes in their districts in subsequent years? *<u>Recommendation:</u> The literature seems to suggest that some type of feedback system (i.e., computerized, diagnostic feedback, expert consultant (in or outside of district, and in person or via computer recommendation), etc.) appears to be more helpful for teachers*

	• Stecker & Fuchs (2000 *M*) Students who received individualized instruction modified based on teacher analysis of their CBM data performed significantly better than did their partners (teachers pre-judged partners to have equal skills and provided same instructional modifications but did not see CBM data on partner student). Teacher Self-Reflection/Self-Regulation • Allinder & Beckbest (1995 *M*) Teacher self-monitoring (use of self-recording and self-eval. questionnaires) resulted in effective teacher compliance to use of CBM • Allinder, Bolling, Oats & Gagnon (2000 *M*) Use of CBM plus teacher self-monitoring resulted in more instructional changes & skills targeted and SPED students made significantly greater progress than teachers who used CBM alone or control group Other • Graney & Shinn (2005 *R*) Neither Group nor Individual feedback had effect on achievement; GE teachers need more than a graph with a slope estimate Variations in Goal Structure • Fuchs, Fuchs & Hamlett (1989c *M*) Dynamic goal CBM resulted in frequent increases of goals and set ambitious goals than static goal CBM (teachers) and showed reliable gains in math than control (students). No reliable differences on content coverage across conditions.	
Qualitative indicators of writing performance	• Coker (2006 *W*) – student background, literacy skill, 1st grade teacher, and 1st grade classroom environment were predictors of writing • Coker & Ritchey (2010 *W*) – qualitative score stronger validity for young students suggesting production scores less appropriate for evaluating young students; article also critically analyzes the 6+1 Traits indicating a lack of research and questions the differences predicted by each trait • Tindal & Hasbrouck (1991 *W*) – seeks to describe "a system for developing diagnostic information" in early writing; employs 3 rubrics • Sturm, Cali, Nelson & Staskowski (2012 *W*) – details 14 point ordinal scale that qualitatively captures developmental states of writing • Richey & Coker (2013 *W*) - validity of picture story scored by qualitative rubric was .44. A smaller increase of slope estimate in the qualitative score for picture story.	*<u>Recommendation:</u> Despite differences in the scope and topic of these articles they seem to suggest that qualitative indicators of writing may yield important information regarding students' early writing performance.*

NOTE: R = Reading, S = Spelling, W = Writing, M = Math

Selecting and implementing interventions.
The Writing Intervention group completed the following tasks:

a) Reviewed literature resources already gathered from the grant proposal and conducted a literature search to find additional articles related to the area of writing instruction and intervention. We focused on research that would assist us in determining effective writing instruction and intervention in the primary grades.

b) Conducted a synthesis of the literature meeting the following criteria:
   - the study described implementation of a writing intervention in the area of handwriting, spelling, text generation, and/or self-regulation,
   - students were in kindergarten through third grade and identified as at-risk in writing or having writing difficulties,
   - at least one dependent variable measured composition in terms of length and quality,
   - the study applied experimental, quasi-experimental, or single-case designs examining causal or functional relations between writing performance and writing interventions, and
   - the study was published in English in a peer-reviewed journal in order to capture studies meeting a refereed standard of quality.

   A total of 29 articles were identified for inclusion in this literature synthesis. Research identified was evaluated based on the effectiveness of the interventions (e.g., effect size) as well as the quality of each study as measured by quality indicators suggested by Gersten et al. (2005) and Horner et al. (2005). We plan to use the results of the effectiveness of the studies in constructing teaching guidelines and/or lesson plans as research-based practices. Appendix E provides a summary of this literature review.

c) Drafted a template for the basic Writing Intervention Plan, scripted activities for Transcription, and developed materials for Transcription. Examples are provided in Appendix F. These materials are being piloted as part of a student dissertation. We are developing additional lesson materials for Text Generation and Self-Regulation

d) Met with Dr. Steve Graham (during a conference in February) to obtain initial feedback and resources related to writing assessment and intervention.

3. **Develop fidelity of implementation instrument.** The Accuracy of Implementation Rating Scale (AIRS) is under revision. See Appendix G for draft.

4. **Develop learning modules for DBI-TLC.** We are developing the learning modules for DBI-TLC to align with the objectives in the DBI manual. The modules will focus on each of the major DBI components listed in Table 2. To develop these modules, we first discussed our overall professional development (PD) plan with Content Expert Stephanie Al Otaiba. We are currently designing a framework, master plan, and templates for each PD module; content for each module will then be developed and inserted into the templates, along with corresponding materials.

5. **Develop teacher knowledge and skills assessment.** This assessment will be designed in April-May, 2014, once the learning modules are complete.

6. **Host summer institute for implementers.** To be conducted in July (CPS) and August (MPS) of Summer, 2014.

C. What opportunities for training and professional development has the project provided? Thus far, GRAs on the project have received training and experience in the following areas:

- CBM-Writing administration, scoring, and interrater agreement procedures.
- Development of tools related to writing assessment and intervention.
- Cross-site collaboration.
- Synthesizing literature and conducting article reviews

D. How have the results been disseminated to communities of interest? Nothing to report, given that we are currently in the initial development phases of the project.

E. What do you plan to do during the next reporting period to accomplish the goals? Table 4 provides a summary of activities to be completed in the next reporting period.

Table 4
Project Activities and Status

Project Activity (As listed in Performance Agreement)	Status
Identify implementers.	In progress; to be completed by 5/14
Develop and revise the DBI protocol (Manual with processes and tools).	In progress; to be completed by 5/14
Develop fidelity of implementation instrument.	In progress; to be completed by 5/14
Develop learning modules for DBI-TLC.	In progress; to be completed by 5/14
Develop teacher knowledge and skills assessment.	To begin in April; to be completed by 5/14
Host summer institute for implementers.	To be conducted in July-August, 2014
Plan collaborative support component of PD system with School Partners and Content Experts.	To begin in Summer, 2014
Develop and collect data on collaborative support activities.	To be conducted during Year 2 (2014-15)
Make final revisions to activities, using input from School Partners and Content Experts.	To be conducted during Year 2 (2014-15)
Conduct feasibility test of DBI-TLC, then revise DBI-TLC.	To be conducted during Year 2 (2014-15)

PRODUCTS

A. Publications and Presentations (none at this time)

B. Website (none at this time)

C. Other Products

- Appendix A: DBI Manual
- Appendix B: CBM-W Administration and Scoring Procedures (revised)
- Appendix C: Decision-Making Rubric
- Appendix D: Change in Instructional Plan form
- Appendix E: Early Writing Intervention Literature Review
- Appendix F: Sample Writing Intervention Plan
- Appendix G: Accuracy of Implementation Rating Scale (AIRS)

PARTICIPANTS AND OTHER COLLABORATING ORGANIZATIONS

A. What individuals have worked on the project?

Name	Project Role	Person mo.	Contribution	Funding Support	Foreign Country
Kristen McMaster	Co-Principal Investigator	1.2	Oversees all aspects of project; coordinates with Content Experts and School Partners	IES	NA
Erica Lembke	Co-PI	1.2	Oversees all aspects of project; coordinates with Content Experts and School Partners	IES	NA
Ted Christ	Co-Investigator	0.6	Provides leadership in assessment aspects of the project	IES	NA
Bonita Janda	MN Project Coordinator	0.8		IES	NA
Kim Moore	MO Project Coordinator	2.25		IES	NA
Carol Garman	MO Project Coordinator	2.25		IES	
Dana Brandes	MN Graduate Research Assistant	3.0	Helps coordinate project, completes development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Amy Kunkel	MN Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Pyung-Gang Jung	MN Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Jaehyun Shin	MN Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Michael Herriges	MN Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Abby Carlisle	MO Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Apryl Poch	MO Graduate Research Assistant	1.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA
Aaron Moss	MO Graduate Research Assistant	0.5	Completes literature review and development tasks (assessment materials, data-based decision-making, & intervention procedures)	IES	NA

What other organizations have been involved as partners?
Minneapolis Public Schools, MN and Columbia Public Schools, MO are serving as the district sites for all school-based activities described in this report.

Have other collaborators or contacts been involved? No.

IMPACT

A. Impact on the development of the principal disciplines of the project

The current project has the *potential* to impact the principal disciplines (writing, special education) in the following ways:

- Improve struggling writers' outcomes in writing.
- Improve teachers' administration, scoring, and use of CBM in writing.
- Improve teachers' knowledge, skills, and self-efficacy related to DBI.
- Facilitate teachers' implementation of DBI.
- Facilitate teachers' capacity to provide instruction targeting multiple dimensions of writing.
- Contribute to knowledge in the field regarding how to provide PD and support related to teachers' implementation of DBI in writing.

B. Impact on other disciplines (None at this time.)

C. Impact on the development of human resources

This project has provided opportunities for graduate research assistants and teachers to learn about theoretical models of writing, assessment and intervention for beginning writers, intervention design, research design, and to gain experience developing and piloting a variety of assessments and intervention protocols.

D. Impact on resources that form infrastructure

None at this time.

E. Impact on technology transfer

None at this time.

F. Impact on society

None at this time.

G. Dollar amount spent in foreign countries

NA

CHANGES/PROBLEMS

There are no changes or problems to report.

BUDGETARY INFORMATION

Below is a description of project expenses through 2/28/14.

Principal Investigator Kristen McMaster, Ph.D. (1.2 person months)
Dr. McMaster has been collaborating with Dr. Lembke to oversee all major grant activities, including budget management; supervision of project coordinator and GRAs at their respective sites; study design and planning; development and revision of all intervention components; coordinating with School Partners for planning, recruiting, and conducting research in schools; obtaining feedback from School Partners, Content Experts, and DBI Implementers; conducting the Summer Institute and other training meetings; overseeing school-based study activities; data analysis and interpretation; writing reports; and disseminating findings.

Co-Investigator Theodore Christ, Ph.D. (.60 person months)
Dr. Christ participates in study design and planning,; development and revision of intervention components and Teacher Knowledge and Skills assessment, and meetings with School Partners, Content Experts, and DBI Implementers.

Project Coordinator Bonita Janda. (.80 person months)
The Project Coordinator is responsible for coordinating and carrying out ongoing project activities, including training and supervising GRAs; scheduling, coordinating, and attending all meetings; communicating regularly with co-PIs, GRAs, School Partners, and Content Experts; coordinating Summer Institutes and other training meetings; coordinating school-based study activities; coordinating data coding, entry, and analysis; assist in writing reports and disseminating results.

Graduate Research Assistant (3.0 person months)
One GRA assists with study planning, development of materials, recruitment, participating in all meetings, data collection and coding/scoring/entry, and conducting classroom consultation visits and observations. She attends weekly project meetings to provide PI and Co-I with updates and to engage in any problem solving needed to provide technical assistance to teachers. In addition, this GRA assists the project coordinator, whose time had to be decreased for the first year of the project (described below under Changes to Budget).

Three Graduate Research Assistants (GRA) (6.0 person months)
Three GRAs assist with study planning, development of materials, recruitment, participating in all meetings, data collection and coding/scoring/entry, and conducting classroom consultation visits and observations. GRAs attend weekly project meetings to provide PI and Co-I with updates and to engage in any problem solving needed to provide technical assistance to teachers.

Fringe Benefits
Fringe benefits are calculated at the University of Minnesota academic rate of 34.9% for PI, Co-I and Project Coordinator. Graduate student fringe benefits are calculated at 16.76% plus $16.82 per hour tuition remission.

Materials and Supplies ($28)
So far, only minimal supplies (e.g., resource materials for intervention development, pencils for scoring CBM probes) have been needed to complete development activities.

Content Experts Stanley Deno, Ph.D., Lynn Fuchs, Ph.D., Steve Graham, Ph.D., and Stephanie Al Otaiba, Ph.D.; @ $1,000/day in Yr 1
Thus far, only one content expert (Stan Deno) has invoiced us for consulting activities. We are prompting Drs. Fuchs, Graham, and Al Otaiba to invoice us for activities completed so far.

Photocopying ($14)
Duplicating costs are for proofing and testing development materials.

Telecommunications ($43)
Telecommunications costs are due to teleconferences with School Partners, Content Experts, and cross-site training activities (conducted via Adobe Connect and conference call).

Non-capital equipment ($1,048.00)
A Dell computer and monitor were purchased to complete project activities.

Changes to the Budget
The following minor changes have been made to the budget:

Reduction in Project Coordinator's FTE. Project Coordinator Bonita Janda is currently working on another grant and was unable to increase her time to the proposed level (6 person months) for Year 1. She will increase to this level in Year 2. In the meantime, I hired an additional GRA to help with project coordination activities.

Change in Content Experts. We had originally designated Dr. Janette Klingner to be the Professional Development expert. Unfortunately, Dr. Klingner recently passed away. Dr. Stephanie Al Otaiba (Southern Methodist University) has agreed to serve as the PD expert.

Also, please note that expenses have been minimal thus far, given that the first half of Year 1 has been devoted to development activities requiring few materials, etc. Also, our subcontractor (University of Missouri) did not invoice the University of Minnesota in time to be included in this report. Expenses for Year 1 are expected to accrue as proposed in the latter half of Year 1.